Scientific Computing on Hardware

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***Abstract*—In this paper, various algebraic and transcendental functions are implemented on microcontroller hardware. This is done by first modeling most functions using high school level differential equations, converting them to difference equations using elementary numerical techniques and then porting these algorithms to an ATMEGA328P microcontroller using AVR-GCC. In the process, we compare different numerical methods and finalize the Runge-Kutta (RK4) method for hardware implementation, based on numerical accuracy. We demonstrate the practical utility of this approach by building a scientific calculator based on the RK4 algorithm.**

***Index Terms*—Scientific Calculator, Microcontroller, Runge-Kutta Method, Quake III Algorithm, Shunting Yard Algorithm, Embedded C.**

I. Introduction

Microcontrollers often find application as the bridge between physical systems and discrete digital logic. This can be as control systems for robotic applications or state estimation for autonomous navigation, among many other applications.[1] Thus, digitally realising continuous-time dynamic systems is an important task in embedded systems engineering. One important aspect of digitising physical systems is the solving of differential equations describing the state. This requires the design and implementation of algorithms capable of approximating complex functions under tight resource constraints. With respect to the ubiquity of differential equations as descriptors of real–world systems, the algorithms under the lens are those that perform or approximate numerical integration.[2]

To evaluate the superiority of any numerical method, one must first consider the fundamental challenge of the Initial Value Problem (IVP). Physical systems are typically modeled as a system of ordinary differential equations (ODEs) where the rate of change of a state variable y is a function of time t and the state itself: $\frac{dy}{dt} = f(t, y)$, given an initial condition $y(t_0) = y_0$. The numerical solution involves advancing the state from a known time t_n to a future time t_{n+1} separated by a step size h. The accuracy of this transition is mathematically governed by the Taylor series expansion. Numerical methods differ in how many terms of the expansion of the true solution $y(t + h)$ they capture.[2]

Numerical methods for function approximation on hardware often prioritize either simplicity or specialized efficiency. Euler's method is the simplest to program, yet its first-order accuracy ($O(h)$) leads to significant numerical drift and instability unless excessively small step sizes are used. Second-order Runge-Kutta (RK2) methods like Heun's improve precision but still feature much greater absolute error magnitudes than RK4.[Appendix B]

Alternative approximation strategies such as Taylor series or Maclaurin expansions are geometrically intuitive but often impractical for embedded systems.[3] This is because these methods require either the tedious recursive calculation of high-order derivatives, which are computationally expensive to evaluate, or for the coefficients for each function to be pre-stored, which results in memory usage piling up quickly for acceptable accuracy.[4] Also, the number of terms, and thus calculations needed for convergence, are much higher for Taylor series as compared to RK4 for the same accuracy.[5] In contrast, the RK4 framework achieves high-order accuracy without needing derivatives, instead evaluating the function at four strategically chosen intermediate points. This allows for a more robust and maintainable implementation when derivatives are complex or unavailable.[6]

For specific transcendental functions, the CORDIC algorithm is highly efficient as it uses simple shift-and-add operations instead of multiplications.[7] However, CORDIC is limited by specific convergence ranges and requires different recursive sequences for different classes of functions.[8] Furthermore, linear multistep methods like Adams-Bashforth can reduce function evaluations but are not self-starting and necessitate a queue of past history to be stored that increases memory usage.[9] The single-step nature of RK4 is more suitable for limited microcontrollers as it avoids the RAM overhead of managing past solution data.

More advanced structural equations and orthogonal collocation methods offer low levels of error but frequently involve complex and heavy computations or iterative non-linear solvers.[10] Similarly, adaptive methods like ode45 (used in MATLAB) provide excellent accuracy but introduce variable execution times that can cause unacceptable jitter in real-time environments, such as the those mentioned above.[1] A fixed-step RK4 approach is superior for predictable performance under the constraints discussed in this work, ensuring that calculations complete reliably within strict sampling windows, such as those in robotics or flight controllers.[6]

Despite extensive research into specialized hardware like FPGAs or complex applications such as GPS orbit computation and motor control, there is a lack of a mathematically unified framework for general scientific computing on 8-bit platforms.[11], [12], [1] Most related works treat different transcendental functions as separate implementation problems rather than a single unified mathematical structure.[13] This

work bridges that gap by reformulating diverse scientific functions into high school-level ODEs, providing a robust, high-precision, and conceptually accessible framework for relatively limited microcontrollers like the ATMEGA328P.

II. Software Implementation

This paper presents the implementation of scientific calculator functions such as trigonometric, logarithmic, exponential, power, factorial, and expression parsing using the C programming language on a microcontroller platform. Various numerical algorithms such as Runge–Kutta (RK4), series expansions, iterative methods, and parsing algorithms have been employed. Each function is discussed with its algorithmic basis followed by its C implementation. Instead of using "general expansion methods" (like Taylor/Maclaurin series with many terms), each function was reformulated as a differential equation. The algorithms used to calculate the values are RK4 (for ODEs), Newton-–Raphson (for refinements like inverse sqrt).

A. Supported Functions

TABLE I
Supported Functions and Operations

Category	Functions / Operations
Trigonometric	$\sin(x), \cos(x), \tan(x)$
Inverse Trigonometric	$\arcsin(x), \arccos(x), \arctan(x)$
Exponential / Logarithmic	$e^x, \ln(x)$
Power / Root	x^y, $\frac{1}{\sqrt{x}}$
Factorial	$n!$
Basic Arithmetic	$+, -, \times, \div$
Constants	π, e
Input Symbols	Digits 0–9, decimal point, parentheses $(,)$

B. Runge–Kutta Method (RK4)

The fourth-order Runge–Kutta (RK4) method is employed to solve ordinary differential equations for trigonometric and exponential functions.[14], [15] This was chosen over alternative ODE solvers due to maintainability and accuracy. A detailed analysis of the choice of algorithm is given in Appendix B. The full derivation and step-by-step procedure for RK4 are provided in Appendix A.

C. Trigonometric Functions

1) Sine Function: The sine function is computed by solving the second-order differential equation: [16]

$$\frac{d^2y}{dx^2} + y = 0, \quad y(0) = 0, \quad y'(0) = 1 \tag{1}$$

using the fourth-order Runge–Kutta method.

System:

$$\frac{dy}{dx} = z, \quad \frac{dz}{dx} = -y, \quad y(0) = 0,\ z(0) = 1 \tag{2}$$

The k_1, k_2, k_3, k_4 correspond to the RK-4 terms of the equation $\frac{dy}{dx} = z$ and the l_1, l_2, l_3, l_4 corresponds to the RK-4 terms of the equation $\frac{dz}{dx} = -y$

RK4 step values:

$$k_1 = hz_n, \tag{3}$$
$$l_1 = -hy_n, \tag{4}$$
$$k_2 = h\left(z_n + \tfrac{l_1}{2}\right), \tag{5}$$
$$l_2 = -h\left(y_n + \tfrac{k_1}{2}\right), \tag{6}$$
$$k_3 = h\left(z_n + \tfrac{l_2}{2}\right), \tag{7}$$
$$l_3 = -h\left(y_n + \tfrac{k_2}{2}\right), \tag{8}$$
$$k_4 = h(z_n + l_3), \tag{9}$$
$$l_4 = -h(y_n + k_3), \tag{10}$$
$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4), \tag{11}$$
$$z_{n+1} = z_n + \tfrac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4). \tag{12}$$

2) Cosine Function: The cosine function is computed by $\sin(\frac{\pi}{2} - x)$.[16]

3) Tangent Function: Tangent is implemented as the ratio of sine and cosine.[16]

D. Inverse Square Root

1) Inverse Square Root: The fast inverse square root algorithm (Quake III method) is used to efficiently compute $x^{-1/2}$. Detailed derivation and Newton-–Raphson refinement are included in Appendix A.

E. Inverse Trigonometric Functions

1) Arcsine: The ODE for the arcsine function is ODE: [16]

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}, \quad y(0) = 0 \tag{13}$$

2) Arccosine: The identity used is: [16]

$$\arccos(x) = \frac{\pi}{2} - \arcsin(x) \tag{14}$$

3) Arctangent: The ODE for the arctan function is ODE: [16]

$$\frac{dy}{dx} = \frac{1}{1 + x^2}, \quad y(0) = 0 \tag{15}$$

F. Logarithmic Functions

The natural logarithm can be obtained by solving the differential equation: [16]

$$\frac{dy}{dx} = \frac{1}{x}, \quad y(1) = 0 \tag{16}$$

which has the exact solution $y(x) = \ln(x)$. Using the RK4 method, $\ln(x)$ can be approximated by integrating this ODE from $x = 1$ to the target value.

G. Exponential Functions

1) Power Function: The power function $y = x^w$ satisfies: [16]

$$\frac{dy}{dx} = \frac{w}{x}y, \quad y(1) = 1 \tag{17}$$

H. Factorial Function

The algorithm is a simple recursive function in C:

$$y_n = n \cdot y_{n-1} \tag{18}$$

[17]

Algorithm 1 Recursive Factorial Function

```
function FACTORIALRECURSIVE(x)
    if x ≤ 1 then
        return 1
    else
        return x × FactorialRecursive(x − 1)
    end if
end function
```

I. Expression Parsing

The Shunting Yard algorithm is employed for converting infix expressions into postfix form for evaluation.

J. Overview

The **Shunting Yard Algorithm**, proposed by Edsger Dijkstra, is a method for converting mathematical expressions in infix notation into postfix notation (Reverse Polish Notation, RPN). It relies on two structures:

- A **stack** for operators/functions
- An **output queue** for the final postfix expression

K. Algorithm Steps

1) Initialize an empty stack and output queue.
2) For each token:
   a) Numbers → add to output.
   b) Functions → push to stack.
   c) Operators → pop higher/equal precedence operators from stack to output (respecting associativity), then push current operator.
   d) "(" → push to stack.
   e) ")" → pop to output until matching "(" is found, discard brackets.
3) After processing, pop remaining operators to output.

L. Operator Precedence

TABLE III
OPERATOR PRECEDENCE AND ASSOCIATIVITY

Operator	Precedence	Associativity
$\vee$ (power)	4	Right
$*$, /	3	Left
$+$, $-$	2	Left

M. Example

For the expression $3 + 4 * 2/(1 - 5)$, the Shunting Yard Algorithm yields the postfix form:

$$3\ 4\ 2\ *\ 1\ 5\ -\ /\ + \tag{19}$$

Algorithmic Steps :

Algorithm 2 Shunting Yard Algorithm (Infix to Postfix)

```
function SHUNTINGYARD(expr)
    Initialize empty stack operators
    Initialize empty string output
    for each character c in expr do
        if c is a number or '.' then
            Append c to output until number ends
            Append space to output
        else if c is a function then
            Push c onto operators
        else if c = ( then
            Push c onto operators
            Increment unmatched_brackets
        else if c =) then
            while Top of operators ≠ ( do
                Pop from operators and append to output
            end while
            Pop '(' from operators
            Decrement unmatched_brackets
            if Top of operators is a function then
                Pop function and append to output
            end if
        else if c is an operator in {+, −, ∗, /, ^, !} then
            while operators not empty and precedence(Top of operators) ≥ precedence(c) do
                Pop from operators and append to output
            end while
            Push c onto operators
        end if
    end for
    while operators not empty do
        Pop from operators and append to output
    end while
    return output
end function
```

N. Reverse Polish Notation (RPN)

Reverse Polish Notation, also known as postfix notation, is a way of writing mathematical expressions in which operators follow their operands. Unlike infix notation (e.g., $3 + 4$), RPN eliminates the need for parentheses by unambiguously encoding operator precedence.

- Example (infix): $3 + 4$
- Equivalent RPN: $3\ 4\ +$

This simplicity makes RPN especially suitable for evaluation using a stack-based algorithm.

O. Evaluating RPN

Evaluation of RPN uses a stack:

1) Scan tokens left to right.
2) Push numbers on the stack.
3) For operators, pop required operands, apply operation, push result.

For the above example, evaluating the postfix expression gives the result 1 as expected .

TABLE II
RK4 UPDATE EQUATIONS FOR SINGLE-VARIABLE FUNCTIONS

Function	k_1	k_2	k_3	k_4	y_{n+1}
Arcsine $\arcsin(x)$	$\frac{h}{\sqrt{1-x_n^2}}$	$\frac{h}{\sqrt{1-(x_n+\frac{h}{2})^2}}$	$\frac{h}{\sqrt{1-(x_n+\frac{h}{2})^2}}$	$\frac{h}{\sqrt{1-(x_n+h)^2}}$	$y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
Arctangent $\arctan(x)$	$\frac{h}{1+x_n^2}$	$\frac{h}{1+(x_n+\frac{h}{2})^2}$	$\frac{h}{1+(x_n+\frac{h}{2})^2}$	$\frac{h}{1+(x_n+h)^2}$	$y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
Natural Log $\ln(x)$	$\frac{h}{x_n}$	$\frac{h}{x_n+\frac{h}{2}}$	$\frac{h}{x_n+\frac{h}{2}}$	$\frac{h}{x_n+h}$	$y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
Power $y = x^w$	$h \cdot \frac{w}{x_n} y_n$	$h \cdot \frac{w}{x_n+\frac{h}{2}} \left(y_n + \frac{k_1}{2}\right)$	$h \cdot \frac{w}{x_n+\frac{h}{2}} \left(y_n + \frac{k_2}{2}\right)$	$h \cdot \frac{w}{x_n+h} (y_n + k_3)$	$y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

TABLE IV
SHUNTING YARD EXAMPLE FOR $3 + 4 * 2/(1 - 5)$

Token	**Action**	**Stack**	**Output**
3	Add to output		3
+	Push to stack	+	3
4	Add to output	+	3 4
*	Push to stack	+ *	3 4
2	Add to output	+ *	3 4 2
/	Pop *, push /	+ /	3 4 2 *
(	Push to stack	+ / (	3 4 2 *
1	Add to output	+ / (	3 4 2 * 1
-	Push to stack	+ / (-	3 4 2 * 1
5	Add to output	+ / (-	3 4 2 * 1 5
)	Pop until (	+ /	3 4 2 * 1 5 -
End	Pop stack		3 4 2 * 1 5 - / +

P. Complexity and Applications

The algorithm runs in $O(n)$ time, where n is the number of tokens. Combined with RPN evaluation, it provides an efficient method for handling expressions in calculators, compilers, and symbolic computation systems.

Q. Function Handling

The Shunting Yard Algorithm can also be used to implement functions such as $\sin(x)$, $\cos(x)$, and others. Function handling follows these rules:

1) When a function token is encountered, it is pushed onto the operator stack.
2) Arguments are processed as sub-expressions normally.
3) When the closing parenthesis ")" is reached:
   - Operators are popped from the stack to the output queue until the matching "(" is found.
   - The function token itself is then moved from the stack to the output queue.

As a result, function calls are correctly represented in postfix form. For example:

$$\sin(x) \quad \rightarrow \quad x \ \sin, \tag{20}$$

R. Conclusion

The Shunting Yard Algorithm, together with RPN evaluation, ensures efficient parsing and evaluation of mathematical expressions. It respects operator precedence, associativity, and bracket handling, making it a cornerstone in expression processing across computing applications.

S. Code Repository

The complete C implementations of the algorithms discussed in this paper are available at: https://github.com/gadepall/calculator/tree/main/codes. Instructions for installation of a software based prototype for testing the algorithms are given in Appendix C.

```
codes/
|-- ShuntingYard.c
|-- inv_sq_root.c
|-- inv_trig_func.c
|-- log.c
|-- power_func.c
|-- trig_func.c
```

Listing 1. Repository Structure in codes/

III. HARDWARE IMPLEMENTATION

A. Hardware Required

TABLE V
MATERIALS REQUIRED FOR THE SCIENTIFIC CALCULATOR

Quantity	**Component**
25	Pushbuttons
1	LCD 16×2
1	Arduino Uno
–	Jumper wires
1	Potentiometer

- A button matrix for user input.
- An Arduino Uno microcontroller to process inputs and execute calculations.
- A 16×2 LCD connected to Arduino for showing results.
- Connections as shown in Fig. 2.

B. Circuit Connections

TABLE VI
ARDUINO TO LCD PIN CONNECTIONS

Arduino Pin	LCD Pin	LCD Label	Description
GND	1	GND	Ground
5V	2	Vcc	Power Supply
GND	3	Vee	Contrast Control
D2	4	RS	Register Select
GND	5	R/W	Read/Write (fixed to Write)
D3	6	EN	Enable
D4	11	DB4	Data Bus (4-bit mode)
D5	12	DB5	Data Bus (4-bit mode)
D6	13	DB6	Data Bus (4-bit mode)
D7	14	DB7	Data Bus (4-bit mode)
5V	15	LED+	Backlight Anode
GND	16	LED-	Backlight Cathode

C. Button Matrix

The button matrix is a grid of push-buttons arranged in rows and columns. It allows multiple buttons to be connected to the microcontroller using fewer pins.

Working Principle:

- The Arduino scans the matrix by activating each row (setting it LOW) one at a time while reading the columns.
- When a button is pressed, it completes the circuit.
- By identifying the active row and column, the Arduino determines which button was pressed.
- Example: If second row is pressed when first column is active, the first column's second button is pressed.

This reduces the number of pins required to implement a calculator with 25 buttons.

TABLE VII
KEY ASSIGNMENTS IN MODE 1 WITH ARDUINO PIN MAPPING

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	0	1	2	3	4
Row 2	5	6	7	8	9
Row 3	+	-	×	÷	sin(
Row 4	cos(	tan(	$e^($	ln(	Clear
Row 5	Backspace	.	=	Mode shift	π

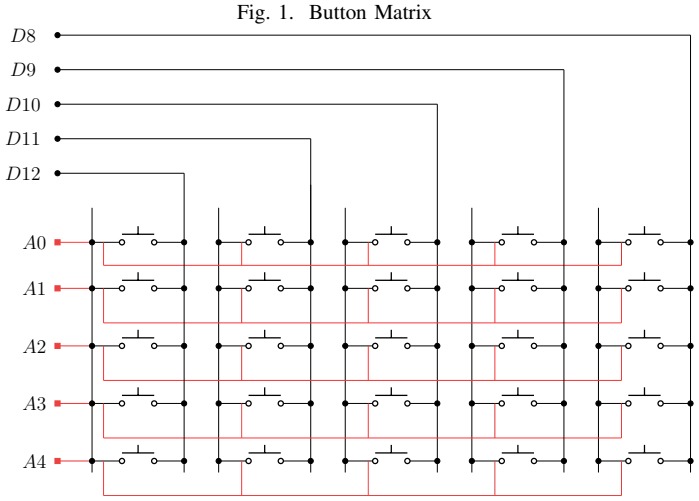
TABLE VIII
KEY ASSIGNMENTS IN MODE 2 WITH ARDUINO PIN MAPPING

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	0	1	2	3	4
Row 2	5	6	7	8	9
Row 3	(	)	x^y	fact(	arcsin(
Row 4	arccos(	arctan(	mod	$\log_{10}($	Clear
Row 5	Backspace	.	=	Mode shift	π

TABLE IX
KEYPAD ROW/COLUMN CONNECTIONS TO ARDUINO PINS

Keypad Line	Arduino Pin
Row 1	D8
Row 2	D9
Row 3	D10
Row 4	D11
Row 5	D12
Col 1	A0
Col 2	A1
Col 3	A2
Col 4	A3
Col 5	A4

Fig. 1. Button Matrix



D. Circuit Schematic

The schematic for circuit connections is shown below,

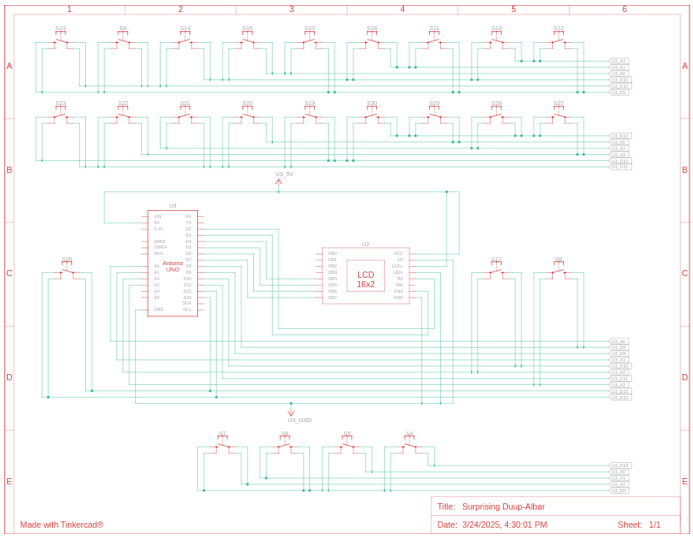


Fig. 2. Schematic of Circuit.

APPENDIX A
RUNGE–KUTTA METHOD (RK4)

The RK4 method is a numerical technique to solve

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \tag{21}$$

with step size h. The next value is calculated as:

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{22}$$

where

$$k_1 = h \cdot f(x_n, y_n), \tag{23}$$
$$k_2 = h \cdot f(x_n + h/2, y_n + k_1/2), \tag{24}$$
$$k_3 = h \cdot f(x_n + h/2, y_n + k_2/2), \tag{25}$$
$$k_4 = h \cdot f(x_n + h, y_n + k_3). \tag{26}$$

Quake III Algorithm: The method obtains an initial approximation by manipulating the IEEE 754 floating-point representation of x, then refines it using Newton–Raphson iterations.

1) Bit manipulation with a "magic constant" ($0x5f3759df$) produces an initial guess y_0.
2) A single Newton–Raphson iteration dramatically improves accuracy.
3) An optional second iteration yields nearly full precision.

Newton–Raphson Refinement: We want to approximate

$$y = x^{-\frac{1}{2}}. \tag{27}$$

Define

$$f(y) = \frac{1}{y^2} - x, \qquad f'(y) = -\frac{2}{y^3}. \tag{28}$$

Applying Newton–Raphson,

$$y_{k+1} = y_k - \frac{f(y_k)}{f'(y_k)} \tag{29}$$

$$= y_k - \frac{\frac{1}{y_k^2} - x}{-\frac{2}{y_k^3}} \tag{30}$$

$$= y_k + \frac{y_k}{2}\left(1 - x y_k^2\right) \tag{31}$$

$$= y_k \left(\tfrac{3}{2} - \tfrac{1}{2} x y_k^2\right). \tag{32}$$

Thus one Newton–Raphson iteration is simply

$$y \leftarrow y\left(1.5 - 0.5\, x\, y^2\right). \tag{33}$$

APPENDIX B
COMPARATIVE ANALYSIS OF NUMERICAL METHODS

Three criteria were considered in the choosing of RK4 as the algorithm for a microcontroller base:

- Computing complexity
- Accuracy for the chosen precision level
- Memory usage

A. Rejected Alternative Algorithms

1) Euler's Method: The Euler method is the simplest numerical solver. The basic idea is to approximate the curve by its tangent line, meaning it is a first order solver.

- **Forward Euler (Explicit)** The slope at the current point is used as the approximation for the value at the end of the step.

$$y_{n+1} = y_n + h \cdot f(x_n, y_n) \tag{34}$$

- **Backward Euler (Implicit)** The slope at the end of the step is used as the approximation. y_{n+1} is present on both sides of the equation, so another algorithm is needed to separate it at each step.

$$y_{n+1} = y_n + h \cdot f(x_{n+1}, y_{n+1}) \tag{35}$$

- – **Drawback:** For a step size of h, the error is $O(h)$. Thus to reduce error, the step size would have to be increased drastically, which is computationally prohibitive.

2) Trapezoidal Rule: This is a second-order method approximates the integral by dividing the area under the curve into trapezoids instead of rectangles.

$$y_{n+1} = y_n + \frac{h}{2} \cdot (f(x_n, y_n) + f(x_{n+1}, y_{n+1})) \tag{36}$$

- – **Drawbacks:** The error is $O(h^2)$. It is also implicit, meaning each step would require another iterative solving mechanism such as Newton–Rhapson to separate the target variable.

Aside: Signal Processing Perspective: Another way to approach the previous formula is tranforming it through the s-domain and z-domain in succession, this reduces the work in solving of ODEs to simple algebraic manipulations. As an example, presented here is the solution to the *sin* function, whose ODE is:

$$y'' + y = 0 \tag{37}$$

1. **Laplace Transform:** Take the Laplace transform of the ODE (assuming zero initial conditions for the transfer function derivation):

$$s^2 Y(s) + Y(s) = 1 \implies Y(s) = H(s) = \frac{1}{s^2 + 1}$$

2. **Bilinear Substitution:** Apply the Bilinear transform substitution $s \approx \frac{2}{h}\frac{1 - z^{-1}}{1 + z^{-1}}$, where h is the step:

$$H(z) = \frac{1}{\left[\left(\frac{2}{h}\frac{1 - z^{-1}}{1 + z^{-1}}\right)^2 + 1\right]}$$

3. **Algebraic Simplification:** Let $c = 2/h$, $A = c^2 + 1$. Simplify and group by powers of z

$$H(z) = \frac{(1+z^{-1})^2}{c^2(1-z^{-1})^2 + (1+z^{-1})^2} \tag{38}$$

$$H(z) = \frac{1+2z^{-1}+z^{-2}}{c^2(1-2z^{-1}+z^{-2}) + (1+2z^{-1}+z^{-2})} \tag{39}$$

$$\frac{Y(z)}{X(z)} = \frac{1+2z^{-1}+z^{-2}}{(c^2+1) + (2-2c^2)z^{-1} + (c^2+1)z^{-2}} \tag{40}$$

$$Y(z) \cdot \left[1 + \frac{2-2c^2}{A}z^{-1} + z^{-2}\right] \tag{41}$$

$$= X(z) \cdot \frac{1}{A}\left[1 + 2z^{-1} + z^{-2}\right] \tag{42}$$

5. **Difference Equation:** Converting from z-domain to discrete time n (where $z^{-k}Y(z) \to y_{n-k}$):

$$y[n] = \underbrace{\frac{1}{A}(x[n] + 2x[n-1] + x[n-2])}_{\text{Input Terms (Source of Energy)}} \tag{43}$$

$$\underbrace{- \frac{2-2c^2}{A}y[n-1] - y[n-2]}_{\text{Feedback Terms (Oscillation)}} \tag{44}$$

Solving for the current term y_n after the impulse has passed:

$$y_n = \frac{8-2h^2}{4+h^2}y_{n-1} - y_{n-2}$$

3) Series Expansion Methods: Taylor/Maclaurin series can also be used for this purpose.

- – **Drawbacks:** Precision cannot be acheived consistently as the number of terms needed vary with function and also with input value. As the input increases, convergence is slow, and number of terms needed to be stored in memory increases.

4) Second-Order Runge-Kutta Methods (RK2): A step up in accuracy from Euler's method are the second-order Runge-Kutta (RK2) methods. These methods use a preliminary slope calculation to make a more accurate step.

- **Heun's Method:** This method averages the slope at the beginning and end of the interval (using an Euler step as a predictor):

$$\tilde{y}_{n+1} = y_n + h \cdot f(x_n, y_n)$$
$$y_{n+1} = y_n + \tfrac{h}{2}(f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1}))$$

  – **Drawback:** The global error is $O(h^2)$. While this is a significant improvement over Euler's method, it is still much less accurate than the fourth-order RK4 method ($O(h^4)$). Given that RK4 only requires twice the number of function evaluations per step as these RK2 methods (four vs. two), the substantial gain in accuracy was judged as a highly favourable trade-off, as shown in Fig.3.

B. Justification for RK4 Selection

The fixed-step RK4 method was determined to be the optimal choice for this specific hardware implementation based on the following analysis (summarized in Table X). The code for this analysis can be found at: https://github.com/gadepall/calculator/tree/main/analysis:

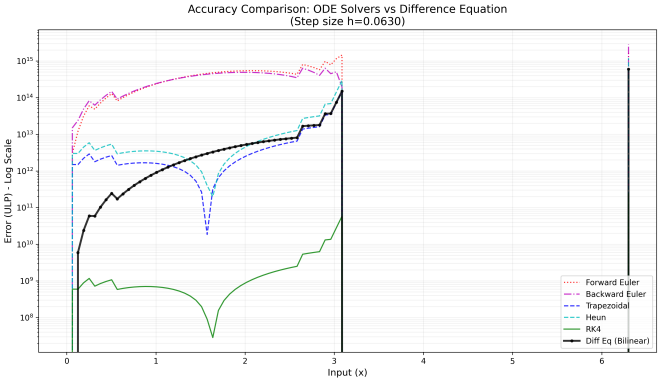


Fig. 3. Comparison of the accuracy of the algorithms

Here, each algorithm was compared against the gold-standard MPFR library up to 50 decimal places. Accuracy is represented using **Units in Last Place (ULP)** measure.

$$\text{Error}_{\text{ULP}} = \frac{|y_{\text{calc}} - y_{\text{true}}|}{\epsilon\,(y_{\text{true}})} \tag{45}$$

As per the results in Fig.3, RK4 demonstrates a clear dominance, with both maximum and average errors being many orders of magnitude less than the other algorithms. The Euler methods demonstrate an increasing drift from the true value with increase in input value, while the trapezoidal method, while stable in its error variation, featured much more absolute error than RK4. As expected, RK2 showed similar error variation as RK4 but much greater in magnitude.

TABLE X
ACCURACY COMPARISON FOR $\sin(x)$ GENERATION ($h \approx 0.06$)

Algorithm	Max Error (ULP)	RMS Error	Stability
RK4	2.38×10^{11}	3.44×10^{-7}	High
Trapezoidal	6.00×10^{14}	8.77×10^{-4}	High
Heun	1.20×10^{15}	1.74×10^{-3}	Moderate
Forward Euler	1.86×10^{15}	8.56×10^{-2}	Unstable
Backward Euler	2.84×10^{15}	7.27×10^{-2}	Unstable

APPENDIX C
SOFTWARE SETUP

A. Termux

1) On your android device, follow the instructions in https://github.com/gadepall/fwc-1 to setup and install Debian on Termux.

B. Platformio

1) Install Packages

```
apt install avra avrdude gcc-avr
    avr-libc
```

2) Follow the instructions in https://docs.platformio.org/en/stable/core/installation/methods/installer-script.html#super-quick-macos-linux to install platformio.
3) Execute the following on debian

```
cd ide/piosetup/codes
pio run
```

See Fig. 4 for the Arduino pin diagram.

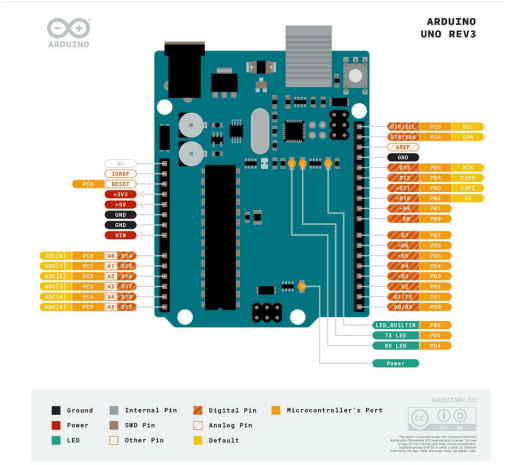


Fig. 4. Arduino Pinout

C. Arduino Droid

1) Install ArduinoDroid from apkpure
2) Open ArduinoDroid and grant all permissions
3) Connect the Arduino to your phone via USB-OTG
4) For flashing the bin files, in ArduinoDroid,

```
Actions->Upload->Upload Precompiled
```

then go to your working directory and select

```
.pio/build/uno/firmware.hex
```

for uploading hex file to the Arduino Uno
5) The LED beside pin 13 will start blinking

D. App Upload

1) Download the files present in https://github.com/gadepall/calculator/tree/main/software.
2) Compile the C files:

```
gcc -shared -o libcalc.so -fPIC func.c
    pars.c
```

3) Copy the app APK file `calci.apk` to the phone Internal Storage and install it from the File Manager.
4) Run the server in `proot-distro`:

```
python3 server.py
```

5) The app now functions as a scientific calculator.

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